



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NSCL Z540-1-1994

ARCONIC TECHNOLOGY CENTER
MATERIALS TESTING
100 Technical Drive
New Kensington, PA 15069
Amy Makovsky Phone: 724 337 2866

CALIBRATION

Valid To: July 31, 2018

Certificate Number: 1019.03

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Chemical

Parameter/Equipment	Range ⁴	CMC ² (±)	Comments
Spark Optical Emission Spectrometers ³	(0 to 0.0005) % (0.0005 to 0.001) % (0.001 to 0.005) % (0.005 to 0.01) % (0.01 to 0.05) % (0.05 to 0.1) % (0.1 to 0.5) % (0.5 to 1) % (1 to 10) % (10 to 50) %	0.0002 % Absolute 0.0003 % Absolute 0.0005 % Absolute 9.8 % Relative 7.9 % Relative 5.1 % Relative 4.0 % Relative 3.0 % Relative 2.2 % Relative 2.2 % Relative	Calibration verification using SRMs. See Table 1 Estimated from 1.0 % to 10 %

II. Dimensional

Parameter/Equipment	Range	CMC ² (±)	Comments
Micrometers	Up to 1.0 in	42 µin	ATCMT SOP-16317-CAL-008
Calipers	Up to 18.0 in	410 µin	ATCMT SOP-16317-CAL-008

III. Mechanical

Parameter/Equipment	Range	CMC ² (±)	Comments
Force – Tension	(10 to 200) lbf (200 to 2000) lbf (2000 to 5000) lbf (5000 to 20 000) lbf (20 000 to 50 000) lbf (50 000 to 100 000) lbf	0.063 lbf 0.6 lbf 2.1 lbf 6.2 lbf 8.6 lbf 59 lbf	ASTM E4, ATCMT SOP-16317-CAL-001
Compression	(10 to 200) lbf (200 to 2000) lbf (2000 to 5000) lbf (5000 to 20 000) lbf (20 000 to 50 000) lbf (50 000 to 100 000) lbf	0.063 lbf 0.6 lbf 2.1 lbf 6.2 lbf 8.6 lbf 59 lbf	



Parameter/Equipment	Range	CMC ² (±)	Comments
Extensometer –			
Gage Length	(0 to 2.0) in	120 µin	ASTM E83, Nikon MeasureScope, ATCMT SOP-16317-CAL-002
Strain	(0 to 1) in	410 µin	ATCMT SOP-16317-CAL-002, ASTM E83, E399, E561, Tinius-Olsen extensometers 111360-5, 118480-2, 115795-1, 139175-4

¹ This laboratory offers commercial calibration service and field calibration services.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ The notation % is defined as weight percent.

Table 1: Typical Maximum Calibrated Concentrations in Weight Percent

Element	Concentration ⁴ (w/o)
Antimony (Sb), Arsenic (As)	0.1 %
Beryllium (Be)	0.3 %
Bismuth (Bi), Lead (Pb)	1.0 %
Boron (B), Cadmium (Cd)	0.05 %
Calcium (Ca)	0.05 %
Chromium (Cr)	0.4 %
Cobalt (Co)	0.02 %
Copper (Cu)	20 %
Gallium (Ga), Phosphorous (P)	0.05 %
Iron (Fe)	3.5 %
Lithium (Li)	4.0 %
Magnesium (Mg)	12 %
Manganese (Mn), Silver (Ag)	1.5 %
Nickel (Ni)	3.0 %
Scandium (Sc), Titanium (Ti)	0.5 %
Silicon (Si)	18 %
Sodium (Na)	0.04 %
Strontium (Sr)	0.1 %
Tin (Sn)	6.0 %
Vanadium (V)	0.2 %
Zinc (Zn)	11 %
Zirconium (Zr)	0.02 %



Accredited Laboratory

A2LA has accredited

ARCONIC TECHNOLOGY CENTER

New Kensington, PA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets R205 – *Specific Requirements: Calibration Laboratory Accreditation Program*. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 26th day of August 2016.

A handwritten signature in black ink, appearing to be 'L. J. ...', written over a horizontal line.

President and CEO
For the Accreditation Council
Certificate Number 1019.03
Valid to July 31, 2018
Revised November 03, 2016

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.